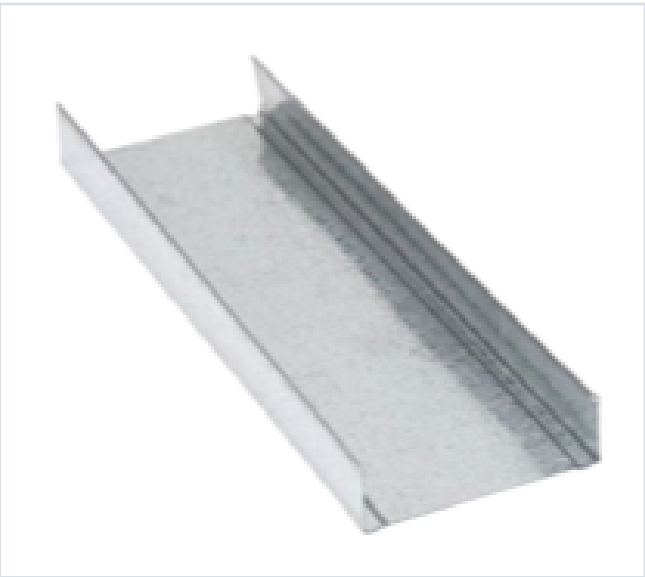


DECKING PRODUCT SUBMITTAL

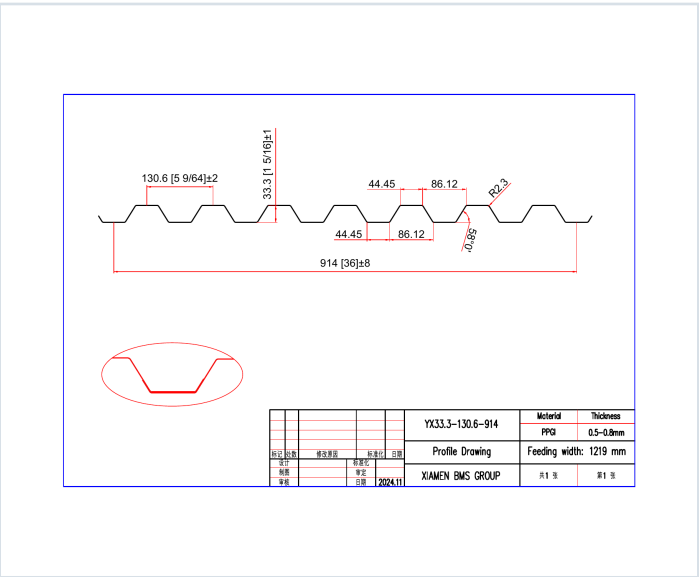
1316FD-FY50-18-G40

PRODUCT	GRADE	GAUGE	COATING
1 5/16" Form Deck	Grade 50	18 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.6	50	0.310	0.294	9291	8812	0.290	0.257

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 70, 71
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

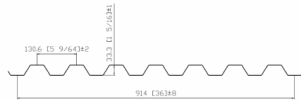
1316FD-FY50-18-G40 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 70

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lb/ft)	E (ksi)	S_x (in ³ /ft)	S_y (in ³ /ft)	ASD ($\phi = 1.67$) M_x (ft-kips/ft)	M_y (ft-kips/ft)	I_x (in ⁴ /ft)	I_y (in ⁴ /ft)
22	0.0096	16	30	0.079	0.070	6308	608	0.007	0.044
20	0.0098	20	30	0.222	0.216	6681	6407	0.010	0.062
18	0.0099	26	30	0.330	0.296	6981	6862	0.016	0.097
16	0.0098	32	30	0.390	0.378	7067	7027	0.023	0.131

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S430-2017, AISI S430-2017 and AISI S430-2018.

Shear and Web Crippling

Gage	V_u (lb/ft)	Web Crippling (ϕ_u) One Flange Loading End Bearing	Web Crippling (ϕ_u) One Flange Loading Interior Bearing
22	2424	1-1/2"	1-1/2"
20	3053	2"	2"
18	5032	3"	3"
16	6299	4"	4"

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S430-2017, AISI S430-2017 and AISI S430-2018.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
Single	22	143	138	133	128	123	118	113	108	103
	20	178	173	168	163	158	153	148	143	138
	18	248	243	238	233	228	223	218	213	208
	16	311	306	301	296	291	286	281	276	271
Double	22	138	133	128	123	118	113	108	103	98
	20	173	168	163	158	153	148	143	138	133
	18	238	233	228	223	218	213	208	203	198
	16	303	298	293	288	283	278	273	268	263
Triple	22	133	128	123	118	113	108	103	98	93
	20	168	163	158	153	148	143	138	133	128
	18	233	228	223	218	213	208	203	198	193
	16	298	293	288	283	278	273	268	263	258

- Note:
- All section properties and ASD ($\phi = 1.67$) uniform loads are calculated in accordance with AISI/S430-2017, AISI S430-2017 and AISI S430-2018.
 - Loads shown in tables are uniformly distributed superimposed loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased to account for other span dimensions.
 - Bending Moment (inches used for design) shall be Single and Two Span $M = \frac{wL^2}{8}$ Three Span or More $M = \frac{wL^2}{10}$
 - Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span condition.

STEEL CATALOG PAGE 71

Span	Single	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
Single	22	68	63	58	53	48	43	38	33	28
	20	86	81	76	71	66	61	56	51	46
	18	106	101	96	91	86	81	76	71	66
	16	128	123	118	113	108	103	98	93	88
Double	22	54	49	44	39	34	29	24	19	14
	20	68	63	58	53	48	43	38	33	28
	18	86	81	76	71	66	61	56	51	46
	16	106	101	96	91	86	81	76	71	66
Triple	22	49	44	39	34	29	24	19	14	9
	20	63	58	53	48	43	38	33	28	23
	18	81	76	71	66	61	56	51	46	41
	16	101	96	91	86	81	76	71	66	61

Note: For loads that cause $L/120$ Deflection, multiply by 2.0. For loads that cause $L/180$ Deflection, multiply by 1.5. For loads that cause $L/360$ Deflection, multiply by 0.67.

Construction Span Table – 20 psf Construction Load

Total Slab Depth	Deck Type	Maximum Unshored Clear Span	1 span	2 span	3 span
150 (1200) 42 PSF	15x6x20 ga	6'-0"	7'-0"	7'-0"	7'-0"
	15x6x18 ga	5'-0"	6'-0"	6'-0"	6'-0"
	15x6x16 ga	4'-0"	5'-0"	5'-0"	5'-0"
400 (1200) 42 PSF	15x6x22 ga	6'-0"	7'-0"	7'-0"	7'-0"
	15x6x20 ga	5'-0"	6'-0"	6'-0"	6'-0"
	15x6x18 ga	4'-0"	5'-0"	5'-0"	5'-0"
	15x6x16 ga	3'-0"	4'-0"	4'-0"	4'-0"
450 (1200) 48 PSF	15x6x22 ga	6'-0"	7'-0"	7'-0"	7'-0"
	15x6x20 ga	5'-0"	6'-0"	6'-0"	6'-0"
	15x6x18 ga	4'-0"	5'-0"	5'-0"	5'-0"
	15x6x16 ga	3'-0"	4'-0"	4'-0"	4'-0"
500 (1200) 54 PSF	15x6x22 ga	5'-0"	6'-0"	6'-0"	6'-0"
	15x6x20 ga	4'-0"	5'-0"	5'-0"	5'-0"
	15x6x18 ga	3'-0"	4'-0"	4'-0"	4'-0"
	15x6x16 ga	2'-0"	3'-0"	3'-0"	3'-0"
550 (1200) 60 PSF	15x6x22 ga	4'-0"	5'-0"	5'-0"	5'-0"
	15x6x20 ga	3'-0"	4'-0"	4'-0"	4'-0"
	15x6x18 ga	2'-0"	3'-0"	3'-0"	3'-0"
	15x6x16 ga	1'-0"	2'-0"	2'-0"	2'-0"
600 (1200) 66 PSF	15x6x22 ga	3'-0"	4'-0"	4'-0"	4'-0"
	15x6x20 ga	2'-0"	3'-0"	3'-0"	3'-0"
	15x6x18 ga	1'-0"	2'-0"	2'-0"	2'-0"
	15x6x16 ga	0'-0"	1'-0"	1'-0"	1'-0"

Note: Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span condition.